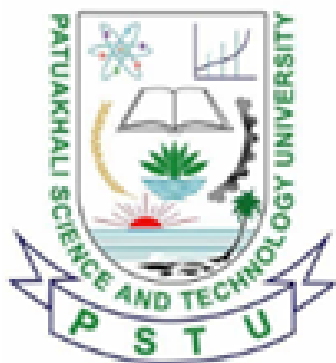




Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional

Project Proposal



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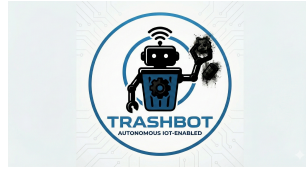
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TrashBot

TRASH COLLECTING ROBOT WITH IOT TECHNOLOGY

1. Introduction

The rapid advancement of robotics offers a transformative solution to the persistent challenge of public sanitation. Manual waste management in densely populated areas is often labor-intensive and inefficient. To address this, we present **TRASHBOT**, an autonomous, IoT-enabled robot designed to streamline waste collection in public spaces.

TRASHBOT integrates autonomous navigation with a robotic arm to detect, collect, and segregate waste. The system features dual operational modes: an **Autonomous Mode** for independent obstacle avoidance and path planning, and a **Manual Mode** for remote operator control via an Android application. By combining mechanical automation with real-time IoT monitoring, TRASHBOT provides a sustainable, cost-effective solution for maintaining cleaner urban environments.

2. Objectives

The primary goal of the **TRASHBOT** project is to modernize public sanitation by developing a cost-effective, autonomous robotics solution that reduces human intervention in waste management.

The specific objectives of this research are:

- **To Design an Autonomous Navigation System:** To develop a mobile rover capable of independent movement, path planning, and obstacle avoidance using ultrasonic sensors and differential steering logic.
- **To Implement Robotic Waste Collection:** To engineer a multi-degree-of-freedom robotic arm driven by high-torque servo motors to detect, grab, and lift trash from the ground.
- **To Enable Smart Segregation:** To integrate metal proximity sensors that automatically distinguish between metallic and non-metallic waste, sorting them into separate internal bins.
- **To Integrate IoT Connectivity:** To allow for dual-mode operation (Autonomous & Manual) and real-time remote monitoring via an Android application (Blynk) using the ESP8266 Wi-Fi module.
- **To Ensure Efficient Power Management:** To design a robust power supply system using high-capacity Li-Ion batteries and DC-DC step-down converters to sustain prolonged field operations.

3. Problem Statement

The development of the TRASHBOT system addresses the following critical deficiencies in current urban waste management practices:

- **Environmental & Public Health Crisis:** The proliferation of uncollected refuse in high-density public areas—such as railway stations and bus terminals—creates severe sanitary risks, serving as a breeding ground for pathogens and contributing to environmental degradation.
- **Operational Inefficiency & Safety Risks:** Conventional manual sanitation is labor-intensive, costly, and inherently inefficient for large-scale coverage. Furthermore, it exposes sanitation workers to hazardous, unhygienic conditions and toxic materials.
- **Technological Gap in Automation:** Current automated cleaning solutions are largely restricted to structured indoor environments. They lack the **autonomous adaptability** required to navigate chaotic outdoor spaces and fail to provide real-time telemetry (IoT) for remote fleet management.
- **Scalability & Consistency Challenges:** Relying solely on human labor makes it difficult to maintain consistent sanitation standards 24/7, particularly in sprawling public infrastructure where continuous monitoring is logistically and financially impractical.

4. Scope

The TRASHBOT project focuses on the design and prototyping of a semi-autonomous rover for dry waste management in public spaces. The core functional scope includes:

- **Dual-Mode Operation:** Implementation of autonomous obstacle avoidance and manual teleoperation via an IoT-enabled Android application.
- **Automated Collection & Segregation:** Development of a robotic arm for physical waste pickup and an inductive sensing mechanism to sort waste into metallic and non-metallic bins.
- **Remote Telemetry:** Integration of real-time video streaming and system status monitoring for remote operators.
- **Prototype Limitations:** The system is optimized for flat terrain and solid dry waste, with a sensor detection range of approximately 30 cm.

5. Methodology

The TRASHBOT project follows an agile prototyping approach, integrating three modular subsystems: **Locomotion**, **Manipulation**, and **IoT Communication**.

5.1. System Architecture

The system relies on a centralized control model driven by a **Raspberry Pi 3 Model B**.

- **Processing & IoT:** The Raspberry Pi executes Python-based control logic, manages video streaming, and transmits telemetry to the Blynk cloud via built-in Wi-Fi.
- **Input/Output:** It processes data from ultrasonic and proximity sensors to drive DC motors (navigation) and servo motors (waste collection).

- **Power:** A high-capacity 11.1V Li-Ion battery with buck converters ensures stable power distribution to all components.

5.2. Technology Stack

Key hardware and software components include:

Category	Component / Technology
Controller	Raspberry Pi 3 Model B (Quad Core 1.2GHz)
OS & Logic	Raspbian OS / Python
Connectivity	Built-in Wi-Fi (802.11n)
Sensors	HC-SR04 (Obstacle), Inductive Proximity (Metal)
Actuation	MG995 Servos & L298N Motor Driver
Power	11.1V 10000mAh Li-Ion, Buck Converters
Interface	Blynk IoT App, IP Webcam
Chassis	4-Wheel Differential Drive Platform

Table 1: Technology Stack of TRASHBOT

6. Budget Estimation

The following is a detailed breakdown of the estimated costs for the hardware components required for the **TRASHBOT** project.

Category	Component Name	Unit Price (BDT)	Qty	Total (BDT)
Controllers	Raspberry Pi 3 Model B (1GB)	10,199	1	10,199
	Raspberry Pi AI Camera	6,880	1	6,880
	ESP-32 Development Board	550	1	550
Motors & Drivers	Gripper	680	1	680
	Tower Pro MG995 servo motor	354	2	708
	Gear Motor with Wheel	155	4	620
	Motor Driver	890	1	890
Sensors	Ultrasonic Sonar Sensor	110	6	660
	Metal Proximity Switch Sensor	390	1	390
	IR LED Sensor	150	1	150
	LED Sensor	99	1	99
	Alcohol Gas Sensor	144	1	144
	Dual IR Flame Sensor	450	1	450
	Water Sensor	350	1	350
	Vibration Sensor	380	1	380
Power	Transformer 12V 3A	390	1	390

	Buzzer	15	1	15
Mechanical & Misc	Full-Size Breadboard	120	2	240
	Half-Size Breadboard	75	2	150
	Jumper Wires (Male-Male)	100	4	400
	Li-ion Battery	260	4	1,040
	Solar Panel	350	1	350
	Battery case	50	1	50
	Display	340	1	340
Module	GSM GPRS Module	1,560	1	1,560
	GPS Module	440	1	440
Grand Total Estimated Cost				28,125 BDT

Table 2: Detailed Budget Estimation

7. Work Plan (Timeline)

The development is divided into phases to ensure the complex mechanical and software systems work together.

Task	Month 1	Month 2	Month 3	Month 4
Mechanical Assembly (Chassis & Drive)	✓			
Circuit Integration & Power Dist.	✓	✓		
Locomotion & Motor Control		✓	✓	
Vision & GPS Integration			✓	✓
Arm Control & Final Testing				✓

Table 3: Gantt Chart for Farm Droid Development

8. Future Plan

With the core autonomous navigation and YOLO-based object detection systems already operational, the next phase of development focuses on scalability, energy efficiency, and ecosystem integration.

8.1. Enhanced Navigation (Lidar & SLAM)

- **Lidar Integration:** Upgrading from ultrasonic sensors to **Lidar (Light Detection and Ranging)** to enable high-resolution mapping.
- **SLAM Implementation:** Implementing **Simultaneous Localization and Mapping (SLAM)** will allow the robot to navigate complex, dynamic environments (like crowded markets) with greater precision than the current differential steering logic.

8.2. Smart City Ecosystem Integration

- **Centralized Dashboard:** Developing a municipal-level dashboard that aggregates data from multiple TRASHBOT units. This would provide city officials with “heat maps” of high-waste zones, allowing for data-driven sanitation scheduling.
- **5G / LoRaWAN Connectivity:** Migrating from Wi-Fi to **LoRaWAN** or **5G** modules to enable long-range telemetry transmission across entire city districts without relying on local hotspots.

8.3. Sustainable Power & Autonomy

- **Solar Hybrid System:** Integrating flexible solar panels on the chassis to extend battery life during daytime operations.
- **Autonomous Docking:** Developing a self-docking charging station where the robot can automatically return and recharge when its battery drops below a critical threshold, ensuring 24/7 continuous operation.

8.4. Swarm Robotics Capabilities

- **Fleet Coordination:** Enabling “Swarm Mode” where multiple TRASHBOT units communicate via a mesh network. If one robot detects an overflow of waste, it can signal nearby units to assist, optimizing cleaning efficiency in large public squares.

9. Conclusion

The **TRASHBOT** project successfully demonstrates the feasibility of an autonomous, IoT-enabled waste management solution designed to address the critical challenges of urban sanitation. By integrating the computational power of the **Raspberry Pi 3 Model B** with robust mechanical actuation and sensor fusion, the system effectively bridges the gap between manual labor and smart automation.

This prototype confirms that automating hazardous waste collection tasks is both technically viable and cost-effective. The implementation of dual-mode operation ensures adaptability in dynamic public spaces, while the intelligent segregation mechanism promotes sustainable recycling practices directly at the source. Ultimately, **TRASHBOT** serves as a foundational step toward smarter, cleaner cities, offering a scalable blueprint for the future of municipal waste management.

The End